Programming in this context is about instructing an Arduino™ about the manner in which it should perform a task. Just as humans need to be trained to be able to carry out an activity, so does the microcontroller. However, the mode of instruction is an altogether different language, a coded form of English.

Without the correct program loaded, an Arduino™ board is like a fruit that's not ripe, i.e. It's not ready to serve its purpose yet. An uploaded program will bring your Arduino™ board to life.

Programs are chunks of coded instructions that tell the microcontroller on the Arduino™ board what task to perform next. Using a series of these coded instructions, the microcontroller performs mathematical operations to make a calculation or take a decision. While programming seems like a difficult task to many people, nothing could be further from the truth.

Programming is only slightly challenging. Look at it like thinking from the point of view of another person. In this case, thinking from the point of view of the microcontroller. To write a program, all we need to do is:

- Think about the task we need to perform.
- Develop a logic and set of mathematical operations that can help perform the task. (Not as hard as it sounds. Using a flowchart to develop the logic can help a lot and is a recommended step for beginners. (Check *http://dgit.in/learnflowchart* for more details on designing a flowchart.)
- Check if the microcontroller can directly use the logic and maths operations.
- In case the microcontroller can't carry out the operation directly, try using alternative ways.
- Tweak the code.

Programming the hardware

If you have prior experience with software programming, you'll notice there are minor differences between software programming and programming a hardware device. Hardware programming is more like writing drivers for your computer hardware.

What is programming?